

Econometrics III
Lindo, Fall 2013
Assignment 2
Due September 11

This is intended to be completed on your own. If you are found to have submitted any code that is not your own, your continuation in the class will be subject to the outcome of a formal student conduct review.

This assignment explores some notable differences between matching and OLS estimators.

PART I: You will begin by considering estimation when there is heterogeneity in the treatment effect and probability of treatment across observable types. (One could think of this as a case in which there is selection on observables but our focus here is not on the bias that omitting this variable will introduce.)

Consider a switching regression model in which $y_{ig} = \beta_g + T_{ig}\delta_g + \epsilon_{ig}$ for $g=1,2$. To keep things simple, set $\beta_1 = \beta_2 = 0$, $\delta_1 = 1$, $\delta_2 = 2$, have $\epsilon \sim N(0, 1)$, and have $Pr[g = 1] = 0.35$. Conduct a series of simulations with 100 replications of $N=1000$ in which $Pr[T_{i2} = 1]$ is held constant at 0.2 but we consider $Pr[T_{i1} = 1]$ from .1 to .9 in intervals of 0.1. As such, you will be conducting 9 different simulations.

For each replication, I want you to retrieve the following: estimates of δ_1 and δ_2 from a *fully saturated* regression model, the estimated average treatment effect based on matching, and the estimated average treatment effect based on a regression model that controls for type. Show how these estimates vary across differing $Pr[T_{i1} = 1]$ by appending the simulation results together and using the command `tabstat <varlist>, by(prtreat1)`. Copy and paste the results of this command into your do file and explain the results (commenting it out so that we can still run the code).

PART II: Now you will consider the relative efficiency of matching and OLS estimators. To limit the number of moving parts, let's remove the heterogeneity in treatment effects and have $\delta_1 = \delta_2 = 1$. Then repeat what you did in PART I but conduct 500 replications for each $Pr[T_{i1} = 1]$. Show that both approaches yield unbiased estimates for all $Pr[T_{i1} = 1]$ with the command `tabstat <varlist>, by(prtreat1)`. Then consider the standard deviation of these estimators across $Pr[T_{i1} = 1]$ with `tabstat <varlist>, by(prtreat1) statistics(sd)`. As in PART I, paste these results into your program and explain what you see.

If your program saves any files, you should include the following to ensure that it runs on my computer and your TAs computer:

```
clear
capture cd "<insert what you want to use as your working directory for this assignment here>"
if c(username)=="jlindo" {
    cd "~/Dropbox/Teaching/Microeconometrics (Grad)/Assignments/Assignment 2 MatchingVsOLS"
}
if c(username)=="apackham" {
    cd "C:\Documents and Settings\apackham\Desktop\ECMT677\Assignment2"
}
```